and astrophysicists are trying to settle this question experimentally, since the answer turns on something that in principle can be measured: the average density of matter in the universe.

If the average matter density exceeds a so-called critical density of about a hundredth of a billionth of a billionth of a billionth (10-29) of a gram per cubic centimeter—about five hydrogen atoms for every cubic meter of the universe—then a large enough gravitational force will permeate the cosmos to halt and reverse the expansion. If the average matter density is less than the critical value, the gravitational attraction will be too weak to stop the expansion, which will continue forever. (Based upon your own observations of the world, you might think that the average mass density of the universe greatly exceeds the critical value. But bear in mind that matter—like money—tends to clump. Using the average mass density of the earth, or the solar system, or even the Milky Way galaxy as an indicator for that of the whole universe would be like using Bill Gates's net worth as an indicator of the average earthling's finances. Just as there are many people whose net worth pales in comparison to that of Bill Gates, thereby diminishing the average enormously, there is a lot of nearly empty space between the galaxies that drastically lowers the overall average matter density.)

By carefully studying the distribution of galaxies throughout space, astronomers can get a pretty good handle on the average amount of visible matter in the universe. This turns out to be significantly less than the critical value. But there is strong evidence, of both theoretical and experimental origin, that the universe is permeated with dark matter. This is matter that does not participate in the processes of nuclear fusion that powers stars and hence does not give off light; it is therefore invisible to the astronomer's telescope. No one has figured out the identity of the dark matter, let alone the precise amount that exists. The fate of our presently expanding universe, therefore, is as yet unclear.

Just for argument's sake, let's assume that the mass density does exceed the critical value and that someday in the distant future the expansion will stop and the universe will begin to collapse upon itself. All galaxies will start to approach one another slowly, and then as time goes by, their speed of approach will increase until they rush together